

LEXFLOW: Workshop on Agentic Workflows for Human-AI Legal Reasoning

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AI is increasingly embedded in legal work, yet current interfaces often expose only final outputs while providing limited support for understanding, supervising, or steering the reasoning process behind them. LEXFLOW frames this as a human-AI interaction challenge, examining how intelligent user interfaces can support process-level supervision of agentic legal reasoning workflows, including workflows involving multiple coordinating AI agents. The workshop brings together researchers from HCI, IUI, AI and Law, and human-centered AI to explore interface designs that make reasoning processes more inspectable, steerable, verifiable, contestable, and accountable. Through paper presentations, co-design activities, and collaborative synthesis, participants will develop reusable design patterns, workflow artifacts, and a shared research agenda for intelligent legal interfaces.

CCS Concepts: • **Human-centered computing** → **Human computer interaction (HCI)**.

Additional Key Words and Phrases: agentic legal AI, human-AI legal reasoning, intelligent user interfaces, legal HCI, multi-agent systems, human oversight, explainable AI, responsible AI

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1 Workshop Overview and Relevance

Legal work is increasingly mediated by AI systems, but current interface design has not kept pace with the judgment, responsibility, and coordination that legal work requires [1, 2, 4, 5]. Across legal research, drafting, compliance, and access-to-justice settings, users are not simply consuming AI outputs but continuously evaluating evidence, interpretations, uncertainty, and whether recommendations can be acted upon. This makes legal AI an important and timely domain for HCI and IUI, where interaction design directly shapes how users understand, supervise, and remain accountable for AI-assisted reasoning [13]. This problem becomes more urgent with *agentic legal reasoning workflows*, where AI-supported legal work unfolds across multiple steps, tools, sources, intermediate decisions, and moments of human intervention, carried out by a single agent or distributed across multiple specialized agents [15, 17].

Current legal AI interfaces remain largely output-centered, often improving efficiency while providing users with limited access to how tasks were decomposed, what information was used, which intermediate choices were made, where uncertainty emerged, or when human input could meaningfully change the result. From an HCI perspective, this

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creates a mismatch between the responsibility placed on users and the interactional resources available to them [11]. The challenge becomes even more complex in multi-agent settings, where reasoning may be distributed across several agents and no single linear path is available for inspection. This makes process visibility and intervention increasingly difficult.

LEXFLOW focuses on the design of intelligent interfaces for agentic legal reasoning workflows, examining how interfaces can support users in inspecting, steering, contesting, and taking responsibility for AI-assisted reasoning by making plans, evidence, uncertainty, and intervention points visible [12]. LEXFLOW brings together researchers across HCI, IUI, AI and Law, and human-centered AI to develop reusable design patterns, evaluation methods, and a shared research agenda for inspectable, steerable, verifiable, contestable, and accountable legal AI.

2 Workshop Organizers

The workshop organizers bring complementary expertise in human AI interaction, legal reasoning systems, and responsible AI. Their recent work on legal fact verification with GenAI studied lawyers' verification practices and identified design requirements for auditable human-AI collaboration in legal practice [5]. Their related work on legal LLM agents and regional legal AI infrastructure further connects LEXFLOW to agentic legal reasoning and legal workflow support [6, 17].

Sirui Han is an Assistant Professor at the Hong Kong University of Science and Technology. Her research focuses on AI, LegalTech, law and finance, and human-centered AI systems. Her recent work examines generative AI in legal fact verification and legal professional workflows. She has also co-organized the ICAIL 2026 workshop *Trustworthy Legal Reasoning: From Generative AI to Verified Legal Intelligence (TLR)*, bringing together researchers in AI and Law, legal reasoning, and trustworthy AI. She will serve as the primary contact and lead the workshop framing, invited-speaker coordination, and community-building strategy.

Yuyao Zhang is a PhD student at the Hong Kong University of Science and Technology working at the intersection of Human-AI Interaction, agentic AI, and intelligent interfaces. Her recent work examines generative AI for legal fact verification and human oversight in legal workflows. She will coordinate submissions, participant communication, and the design activities around agentic legal workflow mapping.

Yidan Huang is a PhD student at the Hong Kong University of Science and Technology whose recent work includes human-centered legal AI and GenAI-supported legal fact verification. She will coordinate reviewing logistics, accepted-paper presentation formats, documentation, and the synthesis of workshop outputs.

3 Relationship to Previous Workshops

At IUI 2026, AI CHAOS! 1st Workshop on the Challenges for Human Oversight of AI Systems [10] and TRUST-CUA: Trustworthy Computer-Using Generalist Agents for Intelligent User Interfaces [8] addressed mixed-initiative oversight and trustworthy AI agents as domain-general interaction challenges. AgentCraft: Workshop on Developing Trustworthy Agentic AI Systems [3] focused on developer-facing tools for building and debugging agentic systems across application domains. At IUI 2025, STEP-HAI [7] examined theoretical and empirical foundations of human-AI interaction. In the broader HCI community, the Workshop on Human-Centered Evaluation and Auditing of Language Models [9, 14, 16] has established an active community around human-centered evaluation of language models and AI agents.

These workshops demonstrate growing interest in human oversight, AI agents, and responsible intelligent interfaces. However, none of them considers legal reasoning workflows as an interaction design problem. Legal work provides a uniquely consequential setting in which users must remain accountable for decisions that depend on AI-supported

reasoning, making it an ideal domain for advancing human-AI interaction research. LEXFLOW complements these efforts by focusing specifically on intelligent interfaces that support process-level supervision of agentic legal reasoning workflows, bringing together the IUI, HCI, AI and Law, and legal AI communities.

4 Participants

LEXFLOW will target 20-30 participants, including authors of accepted papers or demos, invited speakers, organizers, PC members, and researchers from relevant communities. We will solicit 2-4 page position papers and 4-6 page research or demo papers, each reviewed by at least two PC members. Prospective interested participants include Guoying Lu (HKUST), Yanting Zhang (HKUST), Zhizhuo Kou (HKUST), Tuotuo Yang (SJTU), Jielin Feng (FDU), Runhua Zhang (HKUST), Jiayi Zhou (HKUST), Ruyan Chen (CityU), Jiaxiong Hu (HKUST), and Zixuan Zhang (BUAA). We expect them to contribute as paper authors, discussants, demo presenters, or breakout-session participants. Accepted participants will be encouraged to prepare a one-page Workflow Card prior to the workshop. The cards will provide a common format for describing legal tasks, reasoning workflows, evidence sources, AI agents, intervention points, and design challenges, enabling productive discussion across different legal domains and AI systems.

5 Workshop Activities

LEXFLOW will be a full-day workshop combining paper presentations, design activities, and collaborative synthesis. The workshop is organized around hands-on design exercises that help participants externalize and compare agentic legal reasoning workflows. Accepted authors will be invited to prepare a lightweight *Workflow Card* before the workshop, summarizing a legal task, user role, evidence sources, AI agents, uncertainty points, intervention opportunities, and anticipated failure modes. These workflow cards will serve as shared boundary objects throughout the workshop.

09:00–10:00 Opening, participant introductions, and invited talk on human-centered legal AI.

10:00–12:00 Paper and demo lightning presentations, followed by a workflow gallery where participants compare Workflow Cards.

12:00–13:00 Lunch.

13:00–14:30 Co-design Sprint: groups co-design interface concepts for inspectable, steerable, and accountable legal AI workflows.

14:30–15:15 Live Critique: groups present concepts and receive structured feedback from participants.

15:15–17:00 Synthesis of reusable design patterns, evaluation questions, and future collaborations.

6 Planned Outcomes

LEXFLOW will produce a workshop website containing accepted submissions, presentation materials, and post-workshop resources. Beyond a shared research agenda, the workshop will generate reusable community artifacts, including a collection of Workflow Cards documenting diverse legal reasoning workflows and an initial LEXFLOW Pattern Library that captures recurring interface patterns for process visibility, evidence provenance, user intervention, uncertainty communication, multi-agent coordination, and accountability.

The organizers will synthesize workshop discussions into a public report highlighting open research questions, emerging design principles, and evaluation methods for intelligent legal interfaces. These resources are intended to support future research, teaching, and subsequent editions of the workshop while fostering collaboration between the IUI, HCI, AI and Law, and legal AI communities.

7 Tentative Program Committee

The tentative PC will connect HCI, legal AI, AI governance, legal NLP, and lawtech communities. Tentative members include **Yike Guo** (Hong Kong University of Science and Technology), **Xiaojuan Ma** (Hong Kong University of Science and Technology), **Yuhan Luo** (City University of Hong Kong), **Pengcheng An** (Southern University of Science and Technology), **Qian Zhu** (Renmin University of China), **Chuhan Shi** (Southeast University), **Hai-Ning Liang** (The Hong Kong University of Science and Technology (Guangzhou)), and **Mingming Fan** (The Hong Kong University of Science and Technology (Guangzhou)).

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